

Project Scope Definition & User Research Report: Transparent Multi-Objective Food Recommendations Platform

Jan Kuca, Stefan Sidlovsky, Anna Maria Chovancova, and Alvaro Velasco Sobrino

Abstract—This paper presents the scope definition and initial user research phase of a web platform for transparent, multi-objective food recommendations. The platform aims to assist users in selecting food products by simultaneously considering nutritional value and ecological footprint, exposing a fundamental trade-off that existing tools address poorly. We conducted semi-structured interviews to understand current food-purchasing behavior, attitudes toward sustainability, and information needs at the point of purchase. Our analysis reveals that price and convenience dominate purchasing decisions while ecological awareness remains low across most of the sample, and that users strongly distrust opaque scores without explanations but consistently desire personalised recommendations aligned with their individual priorities. These findings motivate a design approach centered on explainable AI (XAI), concrete eco-metric translation, and user-adjustable objective weighting, and directly inform the prototyping phase of the project.

Index Terms—Human-computer interaction, food recommendation systems, explainable AI, sustainable UX, user-centered design, value-action gap, multi-criteria decision support.

I. INTRODUCTION

CONSUMERS today face increasing pressure to make food choices that are both nutritionally sound and ecologically responsible. However, existing digital tools rarely expose the inherent trade-offs between these two objectives, leaving users without actionable guidance at the point of purchase. This gap represents a fundamental Human-Computer Interaction (HCI) challenge: how can an interface communicate complex, multi-dimensional information in a way that is comprehensible, trustworthy, and actionable?

The proposed platform faces four concrete interaction challenges. First, **trade-off visualisation**: users must be able to perceive and resolve conflicts between nutritional and ecological scores without being overwhelmed by data. Second, **explainability**: AI-generated recommendations must be accompanied by reasoning that non-expert users can evaluate and trust, rather than opaque scores. Third, **personalisation without cognitive overhead**: the interface must allow users to express individual priority weights – such as favouring health over environmental impact – through interactions that require minimal effort. Fourth, **bridging the value-action gap**: the platform must translate users’ stated intentions into actionable, effortless purchasing decisions, without resorting to moralising or guilt-inducing framing [3].

This paper reports on the first phase of a user-centered design (UCD) process for a web platform that addresses this

challenge. We begin with a review of related work (Section II), followed by a description of our research methodology (Section III), participant profiles (Section IV), data analysis and resulting artifacts (Section V), and key design insights (Section VI).

II. STATE OF THE ART

Food recommendation platforms that balance nutritional and environmental goals sit at the intersection of three research areas: food recommender systems, Explainable AI (XAI), and eco-nutritional decision support. This section reviews the most relevant works and highlights the gap this project addresses.

A. Explainable Food Recommender Systems

“An Explanation Interface for Healthy Food Recommendations in a Real-Life Workplace Deployment” [4] designs and deploys an explanation interface for a personalised food recommender system, tested across three companies with over 136 participants. The findings show that providing explanations can support healthier eating and improve user trust, but also reveal a tension between users’ immediate preferences and long-term health goals. Importantly, the system only considers nutritional data — environmental impact is not addressed. Our work builds on this by adding an ecological dimension to both the recommendations and their explanations.

B. Semantic Modelling for Food Explanations

“Semantic Modeling for Food Recommendation Explanations” [5] proposes the Food Explanation Ontology (FEO), a post-hoc XAI framework that structures human-readable explanations for food recommendation systems. While FEO offers a solid theoretical foundation, it has no interactive user interface and does not consider environmental metrics. Our project takes the next step by bringing multi-dimensional explanations — covering both health and ecological factors — into a usable web interface.

C. Multi-Objective Recommender Systems

The survey “Multi-Objective Recommender Systems: Survey and Challenges” [6] notes that most published work still optimises for a single metric, missing the complexity of real-world trade-offs. While the algorithmic foundations are well established, the HCI challenge of presenting these trade-offs clearly to non-expert users remains largely unsolved. Our

platform directly targets this gap by designing an interface that communicates health–ecology trade-offs without overwhelming the user.

D. Eco-Aware Food Recommendation

“Knowledge-Powered Recommendation for an Improved Diet Water Footprint” [7] presents a food recommendation system that factors in the water footprint of users’ diets. This is a useful precedent for going beyond carbon as the sole environmental metric. However, it covers only one ecological dimension and provides no explanation of why a product is recommended. Our system aggregates multiple eco-metrics alongside nutritional scoring, and adds an explainability layer so users can understand the reasoning behind each recommendation.

E. Industry Tools: Open Food Facts and Yuka

Open Food Facts [8] is an open, crowdsourced food database offering both a Nutri-Score and an Eco-Score, and has powered over 150 applications including Yuka [1] and FoodVisor. These tools represent the current industry standard for food transparency. However, they display fixed scores without explaining the underlying reasoning, and do not allow users to adjust the relative importance of health versus environmental criteria. No active recommendation engine exists to suggest better alternatives.

F. Research Gap

The reviewed works show a consistent gap: no existing system combines multi-objective food scoring, an alternative-generating recommendation engine, user-adjustable objective weighting, and transparent XAI explanations — all within a single accessible web interface. This project is designed to fill that gap.

III. RESEARCH METHODOLOGY

We employed **semi-structured interviews** as our primary research method. This approach is appropriate for our target population because it combines the structure necessary to ensure coverage of key topics (purchasing behaviour, nutritional awareness, and ecological attitudes) with the flexibility to follow unexpected but relevant threads that emerge during conversation [2].

Interviews were conducted between 8 and 9 May 2026. Six sessions were held in person and two were conducted online via video call. Sessions lasted approximately 25–30 minutes each. Two participants provided explicit consent to audio recording; the remaining six sessions were documented through written notes taken by the interviewer during the conversation. The interview protocol (see Appendix A) was piloted with one participant before the main study and minor wording adjustments were made to two questions that generated confusion during the pilot.

Data was analyzed using **thematic analysis** [2]: two team members independently coded the interview notes, then reconciled their codes through structured discussion to resolve

discrepancies and ensure consistency. The resulting code set was organised into eight recurring themes, which in turn informed the construction of user personas and empathy maps presented in Section V.

IV. PARTICIPANTS

We recruited eight participants through personal networks and word-of-mouth referrals within the university environment. Inclusion criteria required participants to be regular grocery shoppers (at least once per week). We sought diversity in age, household situation, and stated environmental attitudes; all participants reported an omnivore diet, which reflects the convenience-driven food culture of the sample rather than a sampling limitation. Interviews were conducted between 8 and 9 May 2026, in person or via video call, by four team members. No participant names are reported; pseudonymous identifiers assigned during data collection are used throughout. Table I summarizes the participant profiles.

TABLE I
PARTICIPANT PROFILES

ID	Age	Gender	Monthly budget	Eco-awareness
P1	33	M	€400–500	Low
P2	23	F	€100–200	Low
P3	20	F	<€100	Medium
P4	19	F	€400–500	Low
P5	22	M	€100–200	High
P6	42	M	>€500	Low
P7	19	F	€200–300	High
P8	21	F	€100–200	Low

V. DATA ANALYSIS

A. Thematic Synthesis

Analysis of the interview transcripts revealed eight major themes that cut across the full sample of eight participants: (1) diverse but structurally constrained life contexts, (2) price and convenience as default purchasing heuristics, (3) selective and shallow nutritional label engagement, (4) low ecological awareness and its asymmetric distribution, (5) the value-action gap as a systemic barrier, (6) trade-off aversion and nutrition primacy, (7) distrust of opaque scores and AI recommendations, and (8) strong appetite for personalisation. These themes were consistent across participants regardless of age, household composition, or self-reported eco-awareness, suggesting they reflect structural constraints in the food-shopping experience rather than purely individual attitudes.

1) *Theme 1 – Diverse but Structurally Constrained Life Contexts*: The eight participants represented a wide spectrum of life situations, yet their daily food routines were shaped by a common set of structural pressures: time scarcity, budgetary limits, and household responsibilities. Five were students or young adults with monthly food budgets ranging from under €100 to €300, whose shopping habits ranged from daily cooking to relying almost entirely on take-out. Two working adults managed family households with budgets of €400–€500+, one of whom framed all food decisions explicitly around the health of his children. One participant reported

a nut allergy (walnuts and hazelnuts) as an active constraint that must be factored into any recommendation.

This heterogeneity of life contexts carries a direct design implication: the platform must accommodate both the student who shops infrequently in a single discount supermarket and the working parent who plans weekly meals for a family with specific dietary constraints. Any feature that adds friction — mandatory registration, paywalls, complex onboarding — risks immediate abandonment by the most time-constrained and budget-sensitive segments of this population.

2) *Theme 2 – Price and Convenience as Default Purchasing Heuristics*: Price was the most frequently cited decision factor across the sample, functioning in most cases as a hard constraint that other criteria could not override. It was listed among the top factors by seven of the eight participants and, for several of them, it operated not merely as one input among many but as a ceiling that defined the feasible choice set before any other consideration could apply. Participants from lower-budget contexts explicitly connected eco-friendly or high-quality nutritional products to higher costs, creating a perceived incompatibility between responsible purchasing and financial practicality: one participant stated that sustainable alternatives are typically avoided because “*they usually cost more*,” and another articulated that the main barrier to better food choices was simply money

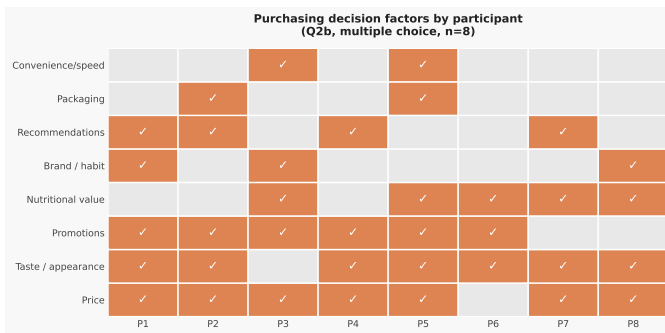


Fig. 1. Purchasing decision factors cited by each participant (Q2b). Price and taste dominate across the sample; nutritional value appears only among participants with higher Q4 scores

Brand familiarity and habit emerged as secondary heuristics, particularly for time-constrained participants who described their shopping trips as largely automatic: they follow a mental list, stick to known products, and rarely deviate unless a promotion or a peer recommendation intervenes. Promotions and discounts were cited as active decision inputs by five participants; taste and visual appearance as tie-breakers by six.

These findings imply that a recommendation platform which requires users to actively seek out information, navigate multiple screens, or manually compare products will face strong behavioural resistance. The interface must reduce, not add to, the cognitive load of the shopping moment, and must find ways to make the better option feel like the cheaper or more convenient one.

3) *Theme 3 – Selective and Shallow Nutritional Label Engagement*: Label-reading behaviour was highly uneven across the sample. Three participants reported never looking

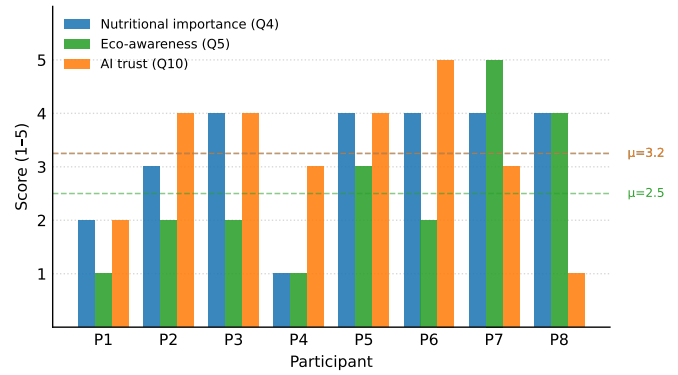


Fig. 2. Per-participant scores for nutritional importance (Q4, $\mu = 3.2$), eco-awareness (Q5, $\mu = 2.5$), and AI trust (Q10) on a 1–5 scale.

at nutritional information on packaging. Four others read labels selectively: one focuses on calorie counts as part of a diet goal; one checks for protein content driven by a fitness objective; one scans for artificial sweeteners and chemicals she actively avoids; and one glances briefly at the nutrition facts panel as a superficial check rather than a systematic comparison. Only two participants had previously tried a food scanning app — one used Yuka and abandoned it because the recommendations did not match her dietary expectations; the other found scanning each product individually with a camera too cumbersome during a real shopping trip.

Despite this shallow engagement with labels, self-reported importance of nutritional value (Q4, scale 1–5) averaged 3.4 across the sample, with five participants rating it at 4 or above. This gap between stated importance and actual label-reading behaviour suggests that the motivation to eat healthily exists but is not being served by current information interfaces. A platform that pre-processes this information and surfaces only the most decision-relevant signals — without requiring the user to scan, decode, or compare raw labels — has clear potential to bridge this gap.

4) *Theme 4 – Low Ecological Awareness and Its Asymmetric Distribution*: Self-reported eco-awareness scores (Q5, scale 1–5) ranged from 1 to 5 across the eight participants, with a mean of approximately 2.4, indicating that ecological considerations are largely absent from everyday food-purchasing decisions in most of the sample. Five participants rated their own eco-awareness at 1 or 2 and reported no specific knowledge of the environmental metrics associated with the food they buy. Two participants stood out with higher scores: one rated herself at 4 and had sometimes changed a purchase for environmental reasons prompted by origin labelling; the most eco-aware participant rated herself at 5 and regularly chose local and seasonal fruit and vegetables — the only participant for whom eco-criteria are a habitual rather than exceptional input.

Notably, none of the participants mentioned carbon footprint, water footprint, or biodiversity impact spontaneously; eco-concerns emerged only when directly prompted by interview questions. This narrow and unevenly distributed awareness has a direct design implication: eco-metrics must be

translated into familiar, concrete comparisons (e.g. “grown 30 km away” or “equivalent to X km driven”) rather than presented as abstract scores, and the platform must not assume any baseline knowledge of environmental indicators on the part of the user.

5) *Theme 5 – The Value-Action Gap as a Systemic Barrier:* A consistent tension emerged between what participants said they valued and what they actually did at the point of purchase. Three distinct types of barrier drove this gap. The first was economic: sustainable and high-quality nutritional products are perceived as more expensive, and in contexts of tight student budgets or family cost management this perception is often accurate. The second was cognitive laziness and time pressure: one participant described herself as “lazy” in the sense of not seeking out information she acknowledged she “could know better,” and several others described shopping as a task to be completed quickly, leaving no room for additional research. The third was motivational detachment: some participants felt that individual food choices have a negligible impact on systemic environmental problems — one stated she was not going to “base her entire diet” on environmental criteria. This rationalisation suggests that the platform’s framing must avoid moralising language or guilt-inducing comparisons, instead presenting eco-improvements as effortless gains that require no lifestyle sacrifice.

6) *Theme 6 – Trade-Off Aversion and Nutrition Primacy:* When presented with hypothetical trade-off scenarios — a product that is healthy but environmentally costly, or one that is eco-friendly but nutritionally inferior — participants consistently resolved the conflict in favour of nutritional value. Six of the eight participants stated that an eco-friendly product with lower nutritional value would not be purchased, preferring the more nutritious alternative regardless of its ecological footprint. Even the participant with the highest eco-awareness score in the sample maintained this position, stating that “nutritional value is still more important, especially when it’s healthy.” Only one participant — also eco-aware — indicated a willingness to prioritise the eco-friendly option in some circumstances. The reverse scenario — a healthy product with a poor ecological record — was treated with greater indifference: most participants stated it would not affect their purchase decision, and three said it would be entirely irrelevant to them.

One participant articulated the underlying reasoning most clearly: “being eco-friendly is important but not at the cost of our health; finding products that have high scores in each section would be the best answer.” This framing — seek the win-win, avoid the trade-off — represents a strong design signal. The platform should never present eco-improvements as coming at the expense of nutritional quality. Trade-offs must be framed as gains (“equally nutritious and better for the environment”) rather than as substitutions (“less protein but lower carbon”). When a genuinely better alternative exists on both dimensions, this must be communicated first; when a real trade-off is unavoidable, the platform must let the user resolve it according to their own priorities rather than imposing a default weighting.

7) *Theme 7 – Distrust of Opaque Scores and AI Recommendations:* General trust in AI recommendations (Q10, scale 1–5) varied widely across the sample, from 1 to 5, with a mean of 3.25. The majority of participants drew a clear distinction between accepting a score and understanding it. Several recounted situations in which they had ignored a recommendation — not because they found it incomprehensible, but because they disagreed with it and lacked the information to evaluate whether the system or their own judgement was correct. One participant captured this precisely: he ignored a recommendation “not because he didn’t understand it, but because he disagreed with it” and the platform offered no way to interrogate the reasoning.

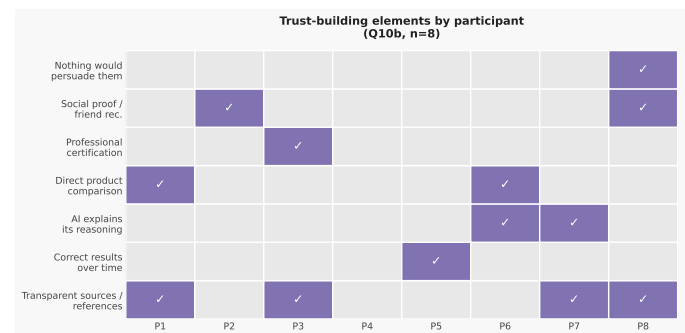


Fig. 3. Elements that would increase trust in AI recommendations (Q10b)

The elements most cited as trust-building were: transparent sourcing of data and references (four participants); direct product comparison as evidence (three participants); professional or institutional certification (one participant); and a low rate of errors over time as accumulated credibility (one participant). Conversely, features that participants said would erode trust or prompt abandonment included: recommendations that conflict with personal taste without explanation; excessive push notifications; information density that is difficult to process quickly; and any monetary barrier to access. One participant with the lowest AI trust score added that she would not be persuaded by any interface improvement — for this user, social proof from real people is the only trusted channel, underscoring that a platform relying entirely on algorithmic authority will not reach all segments of the target population. These findings confirm that explainability is not an optional layer on top of a recommendation engine but a structural requirement for user adoption.

8) *Theme 8 – Strong Appetite for Personalisation:* Despite the diversity of attitudes toward nutrition and sustainability, all eight participants expressed a desire to customise what the platform optimises for. Six answered “yes — definitely” and two “yes — probably” when asked whether they would want to set their own health/eco priority weights via a slider. The specific personalisation goals varied considerably: two participants prioritised high-protein content for fitness-oriented diets; one wanted caloric and sugar targets linked to a weight-management goal; one required that nut allergies be automatically factored into every recommendation; one asked for integration with a specific supermarket’s product catalogue; and one requested that local and seasonal availability be a

first-class filter. Notably, this interest did not correlate with eco-awareness: even participants indifferent to environmental impact wanted control over which criteria the platform optimised for. One participant framed the ideal output as items simultaneously scoring well on eco-friendliness, nutritional quality, and affordability — a triple-optimisation framing that defines the platform’s core value proposition. The priority-weighting slider is a necessary entry point, but the personalisation architecture must accommodate progressive enrichment of the user’s preference profile over time.

B. User Personas

Based on the research data, we constructed five user personas capturing the primary archetypes in our target population. Full persona sheets are included in Appendix B (Figures 4–8).

The Eco-Beginner (Maya Kolsky, 19) is starting an eco-friendly lifestyle but finds complex environmental data overwhelming. Primary pain point: inability to verify eco-claims without seeing concrete sourcing evidence.

The Fitness Pragmatist (Adrien Dawin, 23) prioritises protein content and price above all else. Primary pain point: eco-friendly suggestions consistently ignore protein-to-calorie ratios, making them irrelevant for fitness goals.

The Value-Driven Traditionalist (Simon Joen, 42) shops for a family and wants to understand the reasoning behind recommendations, not just see raw scores. Primary pain point: opaque multi-objective balancing with no allergy integration.

The Traditional Skeptic (Joe Star, 19) is highly price-sensitive and finds unsolicited eco suggestions intrusive. Primary pain point: recommendations that conflict with taste and price without justification.

The Conscious Budgeter (Yun Mo, 22) is motivated by health and transparency but is frequently priced out of sustainable choices. Primary pain point: no tool that balances health, eco-impact, and budget constraints simultaneously.

C. Empathy Maps

Empathy maps for the three primary personas are presented in Appendix B (Figures 9–11). The maps highlight a recurring tension across all participants: while users *think* they should make more sustainable choices, their actual *behavior* defaults to convenience-driven decisions. This aligns with the known **value-action gap** in sustainability research [3].

VI. KEY DESIGN INSIGHTS

From our research, we derive the following insights that will directly guide the prototyping phase:

- 1) **Price and convenience must be treated as hard constraints, not competing values.** Seven of the eight participants identified price as their primary purchasing criterion, and several described it as a ceiling that ruled out all other considerations before any comparison could take place. Habit and brand familiarity further reduced active decision-making during shopping trips. This finding implies that the platform must surface better alternatives without adding cognitive or financial friction:

recommendations must be price-comparable by default, and the interface must require minimal interaction to reach a useful result.

- 2) **Ecological metrics must be grounded in concrete, familiar comparisons.** Self-reported eco-awareness averaged 2.4 out of 5 across the sample, and no participant mentioned carbon footprint, water footprint, or biodiversity impact spontaneously. Eco-concerns emerged only when directly prompted, and focused on the most visible categories such as industrial meat and long-distance transport. Abstract environmental scores will not influence behaviour for this population. The platform must translate eco-metrics into relatable units — for example, “grown 30 km away” or “equivalent to X km driven” — and must not assume any baseline knowledge of environmental indicators.
- 3) **Explainability is a structural requirement, not an optional feature.** General trust in AI recommendations averaged 3.25 out of 5, with wide individual variation. Several participants recounted ignoring a recommendation not because they misunderstood it, but because they disagreed and had no way to interrogate the reasoning. The most frequently cited trust-building elements were transparent data sourcing and direct product comparison as evidence. This confirms that a black-box score — however accurate — will not achieve adoption. Every recommendation must be accompanied by a human-readable explanation of the factors that drove it, with references to verifiable sources.
- 4) **Trade-offs must be framed as gains, never as substitutions.** When presented with hypothetical trade-off scenarios, six of the eight participants resolved the conflict in favour of nutritional value and would not accept an eco-friendly product with lower nutritional quality. The reverse scenario — a nutritious product with a poor ecological record — was largely dismissed as irrelevant. One participant articulated the underlying logic most clearly: “being eco-friendly is important but not at the cost of our health.” The platform must therefore prioritise win-win alternatives — products that score well on both dimensions — and present them first. When a genuine trade-off is unavoidable, it must be framed as a marginal gain (“equally nutritious and better for the environment”) rather than a substitution (“less protein but lower carbon”), and the final choice must always be left to the user.
- 5) **Personalisation is a universal expectation across all user profiles.** All eight participants expressed a desire to control what the platform optimises for, regardless of their level of eco-awareness or nutritional literacy. Six answered “yes — definitely” and two “yes — probably” when asked whether they would use a health/eco priority slider. Personalisation goals ranged from protein targets and allergy filters to supermarket-specific catalogues and seasonal availability. This breadth of expectations confirms that a one-size-fits-all recommendation will not sustain engagement. The priority-weighting slider is a necessary entry point, but the personalisation archi-

texture must be designed to accommodate progressive enrichment of the user preference profile over time.

VII. CONCLUSION

This paper has presented the scope definition and user research phase of a transparent, multi-objective food recommendation platform. Through semi-structured interviews with eight participants, we identified five structural findings that directly shape the design of the system: price and convenience function as hard purchasing constraints that recommendations must respect; ecological metrics are largely unfamiliar and must be translated into concrete, relatable comparisons; opaque scores erode trust and make explainability a structural requirement rather than an optional feature; trade-off framing must favour win-win alternatives and leave final choices to the user; and all participants, regardless of eco-awareness, desire personalised objective weighting. These findings confirm the existence of the value-action gap in our target population and reveal specific design requirements — particularly for trade-off visualisation, explainability, and personalisation — that will structure the prototyping phase. Future work will translate these insights into low-fidelity wireframes, followed by a high-fidelity interactive prototype to be evaluated using the System Usability Scale and the Explanation Satisfaction Scale.

ACKNOWLEDGMENTS

The authors used ChatGPT (OpenAI) for structuring ideas and language refinement during the writing process.

REFERENCES

- [1] Yuka, *Yuka – Food & Cosmetic Scanner*, 2024. [Online]. Available: <https://yuka.io>
- [2] J. Lazar, J. H. Feng, and H. Hochheiser, *Research Methods in Human-Computer Interaction*, 2nd ed. Morgan Kaufmann, 2017.
- [3] A. Kollmuss and J. Agyeman, “Mind the gap: Why do people act environmentally and what are the barriers to pro-environmental behavior?,” *Environmental Education Research*, vol. 8, no. 3, pp. 239–260, 2002.
- [4] R. De Croon, D. Segovia-Lizano, P. Finglas, V. Vanden Abeele, and K. Verbert, “An Explanation Interface for Healthy Food Recommendations in a Real-Life Workplace Deployment: User-Centered Design Study”, Feb. 2025
- [5] I. Padhiar, O. W. Seneviratne, S. Chari, D. Gruen, and D. L. McGuinness, “Semantic Modeling for Food Recommendation Explanations”, 2021
- [6] D. Jannach, “Multi-Objective Recommender Systems: Survey and Challenges”, Oct. 2022.
- [7] S. Joshi, F. Ilievski, and J. Pujara, “Knowledge-Powered Recommendation for an Improved Diet Water Footprint,” in *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, 2024
- [8] Open Food Facts, “Open Food Facts – The Free Food Products Database,” 2024. [Online]. Available: <https://world.openfoodfacts.org>

APPENDIX

INTERVIEW PROTOCOL

A. Introduction

Participants were informed of the study's purpose and their right to withdraw at any time without consequences. Two participants provided explicit consent to audio recording; the remaining six sessions were documented through written notes taken by the interviewer. The following statement was read aloud at the start of each session:

"Hi [Participant's Name], thank you for taking the time to speak with us today. My name is [Interviewer's Name], and I am part of a team at UIB designing a web platform that helps people make more informed food choices. We want to understand how you currently shop for food, what information matters to you, and what challenges you face. There are no right or wrong answers — your honest experience is exactly what we need. Would it be okay if we record this session? The recording will only be used internally for analysis and will remain confidential. This will take approximately 25–30 minutes."

B. Background & Shopping Habits

- 1) Can you tell me about yourself — your lifestyle, household situation, and how often you go grocery shopping?
- 2) Where do you usually buy your food — supermarket, local market, online, or a mix?
- 3) Do you have any dietary restrictions, allergies, or specific diets?
- 4) Walk me through your last grocery shopping trip. What did you buy and how did you decide what to put in your basket?
- 5) When choosing between two similar products, what factors matter most to you?
- 6) Do you look at the nutritional information on food packaging? If yes, what do you look at?
- 7) Have you ever used a food app (e.g. Yuka, Open Food Facts, Nutri-Score) to check a product? What was your experience?

C. Nutrition & Ecological Awareness

- 1) On a scale from 1 to 5, how important is nutritional value to you personally when buying food?
- 2) Are there specific nutrients you actively try to get more or less of — for example protein, sugar, or sodium?
- 3) On a scale from 1 to 5, how aware are you of the ecological impact of the food you buy?
- 4) Are there specific food categories where the environmental impact feels most relevant to you — for example meat, dairy, or imported fruit?
- 5) Have you ever changed a purchase because of environmental reasons? What prompted that change?
- 6) You might care about eating healthily and sustainably — but do your actual purchases always reflect that? If not, what are the main barriers?

D. Trade-Offs & Platform Expectations

- 1) Imagine a product is very nutritious but has a high environmental cost. How would that affect your choice?
- 2) Now the opposite: an eco-friendly product with lower nutritional value. What would you do?
- 3) If a platform showed both a health score and an eco score at the same time, would that be helpful or overwhelming? Why?
- 4) If a platform recommended one product over another, what would you need to see to understand and trust that recommendation?
- 5) Would you prefer a one-line plain-language explanation, a visual chart, or a detailed breakdown? Why?
- 6) Would you want to set your own priorities — for example via a slider: 70% health / 30% environment?
- 7) On a scale from 1 to 5, how much do you trust recommendations made by an AI system? What would make you trust it more?
- 8) Which search filters would you actually use when looking for a product — for example high protein, low sugar, low CO₂, local, or organic?

E. Closing

- 1) If this platform existed today, what single feature would make you actually use it regularly?
- 2) Is there anything that would definitely put you off using it?
- 3) Is there anything we have not covered that you think would be important for us to know?

USER PERSONAS AND EMPATHY MAPS

The Eco-Beginner

Name: Maya Kolsky
 Age: 19
 Role: The Eco Shopper Beginner
 Primary Goal: Buying local nutritional products with low sugar percentage. Wants to know if the fruits are in season and local.
 Info: A person who is starting with eco-friendly lifestyle and doesn't want to lose control over their food budget.

Background & Context:

1. Regularly checks sugar and calories of each product. Cares about seasons of fruits and where the product is from.
2. Besides nutritional value and if the product is local cares mostly about the price.
3. Lives with family and when shopping alone has always a clear plan in mind.
4. Since they are a beginner as an eco buyer they might find complex environmental data too overwhelming.

Pain Points & Challenges

1. **Lack of transparency:** Needs to see why each product was recommended to decide if they should buy them.
2. **Verification of Eco-claims:** Needs to see more information instead of just tags that labels the product eco-friendly or bio-friendly.
3. **Not personal:** Wants to have recommendations based on what they care about.
4. **Regulatory compliance:** Struggles to trust 'green' labels without seeing concrete evidence of the supply chain.
5. **Information Overload:** Hard to find a product that suits their values while reading ten different labels.

Needs & Expectations from the XAI Dashboard

1. **Clear explanations:** For the explanation for each product to be understandable and they can be found on the Internet.
2. **Feature importance:** Needs a simple description or ranking based on why this product was specifically recommended to this person.
3. **Personalisation:** Needs an option to specifically choose which information will be shown about each product.
4. **Alternative scenarios:** Wants to see how the environmental score changes if they choose a different brand or a non-local alternative.

Preferred Explanation Methods

1. **Natural Language explanation:** easy to read and understand even when in hurry.
2. **Layered Information:** The basic information is simple and short but obtains points you can open up to look for more information on that topic.
3. **External Data Integration:** Links for sites from which these information are derived to be believable.
4. **Counterfactual explanations:** How much would the score change if the buyer chose a different product.
5. **Decision Flowcharts:** Since they have a "clear plan in mind," a flowchart showing the step-by-step reasoning they will be very effective for a person who wants to see structured information behind each product.

How this persona guides design thinking

1. Ensures that explainability is user-friendly, not just technically correct.
2. Helps define UI elements for the product descriptions.
3. Guides the development of a 'Deep Dive' UI component where users can verify sourcing data through external API links.

The Fitness Pragmatist

Name: Adrien Dawin
 Age: 23
 Role: Pragmatic Shopper
 Primary Goal: Buying groceries for cooking at home.
 Info: An accomplished buyer who doesn't have much time and cares mostly about being effective.

Background & Context:

1. Person who works out and checks how much protein the product contains.
2. Cares mostly about price and their preference.
3. Lives with sister and a partner. When shopping goes straight to products they want to buy.

Pain Points & Challenges:

1. **Lack of transparency:** The sites offer what is wrong and what is right without any proof that can be found.
2. **Not personal:** The sites and apps offer products that are eco-friendly but not usable to said person.
3. **Not usable:** Eco-friendly suggestions often ignore protein-to-calorie ratios, making them irrelevant for fitness goals.
4. **Long search time:** Needs information that is easy to get and look through.
5. **Cognitive Load:** Since they are "lazy to look for information" any friction in the IO will cause them to revert to mass-produced products.

Needs & Expectations from the XAI Dashboard

1. **Quick:** For the app/site to load and shows information quickly while buying.
2. **Personalisation:** Needs an option to specifically choose which information will be shown about each product.
3. **Easy to read:** Should display only short text. Needs to be easy to skim through.

Preferred Explanation Methods

1. **Comparative Bar Charts / Plots:** Simple and easy to read comparison between products that will show different categories, for example "Protein per euro", "Carbon footprint."
2. **Quick-Glance Icons & Color Coding:** Simple visual system that will show if the product is good for the environment and for the health of the buyer.
3. **Feature Importance Plots:** (SHAP/LIME style visualizations) to see how each impact (eco-impact, protein-impact...) was weighted for this recommendation.
4. **AI analyses and recommendation:** Short text that will say the most important information about the product.
5. **"What-if" Analysis:** A slider that would show how a small change in products and price leads to a big change in environmental impact.

How this persona guides design thinking

1. Prioritizes simpler and easier to look through information formats for approach to showing differences between products.
2. Helps define what information should be included mainly to help choosing people who all have different values.
3. Prioritizes a 'Quick-Compare' dashboard layout that minimizes cognitive load during high-speed shopping.
4. This persona forces the design to include a "One-Click Add" feature or a summary view that doesn't require opening new tabs.
5. This persona justifies the use of color-coding (Green/Yellow/Red) to allow for "skimming" rather than reading.

Fig. 4. User Persona 1 – The Eco-Beginner (Maya Kolsky).

Fig. 5. User Persona 2 – The Fitness Pragmatist (Adrien Dawin).

Persona: The Value-Driven Traditionalist

Name: Simon Joen
age: 42
Role: Health-First Buyer
Primary Goal: Buying healthy food.
Info: A person who cares about their health a lot and it show in his shopping.

Background & Context:

- 1. Lives with a family and goes grocery shopping every week. Working at a stable job.
- 2. Buys products from both supermarkets and local markets.
- 3. Mostly looks at expiration dates and nutritional value while buying products.

Pain Points & Challenges

- 1. **High Interaction Cost:** Manual input for every item made the shopping too slow and inefficient.
- 2. **Lack of Contextual Meaning :** Struggles with apps that provide data without any meaning behind them. Need to know "why" behind a comparison rather than just seeing the raw numbers.
- 3. **Opaque Multi-Objective Balancing:** Difficult to manually calculate the trade-offs between protein content, sugar levels and environmental impact for every product.
- 4. **Sustainability Blind Spots:** Difficulty identifying the true environmental footprint of high-impact categories like meat, fruits, and vegetables without deep research.
- 5. **Rigid Recommendation Logic:** Frustrated by systems that do not allow for the input of personal dietary priorities or allergy constraints, leading to irrelevant suggestions.

Needs & Expectations from the XAI Dashboard

- 1. **Side-by-Side Comparison:** Requires a dashboard that displays products in a grid or list with highlighted differences in key metrics.
- 2. **Dynamic Prioritization:** Needs a UI with sliders or a checklist to set "Protocols" (e.g., "Priority: High Protein, Low Sugar, Local").
- 3. **Automated Scanning:** Expects a solution that doesn't require individual photos, perhaps through a "smart list" or barcode integration to save time.

Preferred Explanation Methods

- 1. **Comparative Bar Charts:** Visualizing the "Protein per Euro" or "Carbon Footprint" across three different products simultaneously.
- 2. **Natural Language Summaries:** A brief explanation of the "meaning" behind a comparison (e.g., "Product A is 10% more expensive but contains 5g more protein and is currently in season").
- 3. **Confidence Scores:** Since they trust AI recommendations, they need a "Match Score" showing how well a product fits their specific dietary and eco-priorities.

How This Persona Guides Design Thinking

- 1. This persona ensures that the platform prioritizes **efficiency** (avoiding slow tasks like photography) and **multi-objective transparency** (explaining the balance between health and the environment).

The Traditional Skeptic

Name: Joe Star
Age: 19
Role: The Price-First Skeptic
Primary Goal: Buy preferred foods at the lowest possible price while occasionally improving health choices.

Pain Points & Challenges

- 1. **Irrelevant Alternatives:** Frustrated by apps that suggest "eco-friendly" items that are more expensive or don't match their taste preferences.
- 2. **Intrusive Suggestions:** Finds unsolicited advice about the ecosystem annoying or preachy, leading to "app fatigue."
- 3. **Price Sensitivity:** High resistance to any recommendation that increases the total cost of the shopping basket.
- 4. **Trust Gap:** Suspicious of AI that tries to "push" a certain agenda (like environmentalism) instead of helping them save money.

Needs & Expectations from the XAI Dashboard

- 1. **On-Demand Recommendations:** Only wants to see suggestions when they explicitly ask for them (e.g., a "Show Healthier Options" button).
- 2. **Price Parity:** Recommendations must be within a similar price range or currently on discount.
- 3. **Taste Matching:** Needs the system to understand that they won't switch from a favorite brand unless the alternative is very similar in flavor.

Preferred Explanation Methods

- 1. **Price-Benefit Comparison:** Bar charts showing that the "healthier" version is actually the same price or cheaper per unit.
- 2. **Simple Health Grading:** A clear "Nutrition Grade" (A-E) rather than complex data about CO2 or soil health.
- 3. **"What-if" Budgeting:** Showing how much money they save by switching to a seasonal (and therefore cheaper) healthy product.

How This Persona Guides Design Thinking

- 1. Directly influences the "Settings" menu where users can toggle "Environmental Impact" OFF and "Health/Discounts" ON.
- 2. Ensures that **Price** and **Taste** are the most prominent data points, with "Eco" data hidden in a secondary layer for this specific user group.
- 3. Prevents user churn by ensuring the AI respects the user's lack of interest in the environment while still providing value through health and savings.

Fig. 6. User Persona 3 – The Value-Driven Traditionalist (Simon Joen).

Fig. 7. User Persona 4 – The Traditional Skeptic (Joe Star).

Persona: The Conscious Budgeter

Name: Yun Mo

Age: 22

Role: Value-Driven Health Seeker

Primary Goal: Finding high-quality, chemical-free, and eco-friendly food without exceeding a strict daily budget.

Info: A frequent shopper who is highly motivated by health and transparency but is often forced to choose price over planet.

Pain Points & Challenges

1. **The "Green" Price Premium:** Frustrated by the high cost of organic and eco-friendly products, often feeling "priced out" of a sustainable lifestyle.
2. **Chemical Transparency:** Struggles to identify hidden artificial sweeteners or chemicals behind complex packaging jargon during a quick daily shop.
3. **Information Accessibility:** Finding it tedious to manually check every new product's packaging for nutritional and environmental data.
4. **Lack of Prioritization Tools:** Traditional shopping apps don't allow them to balance their specific health needs against their budget in real-time.

Needs & Expectations from the XAI Dashboard

1. **Health & Eco Scoring:** Explicitly wants to see standardized scores for every product to make instant, informed comparisons.
2. **Dynamic Goal Setting:** Needs interactive UI elements, like sliders, to adjust priorities between health, eco-impact, and price depending on their current budget.
3. **Trend & Seasonality Discovery:** Wants the system to highlight seasonal products, which are often both healthier and more affordable.
4. **Cost-Effective Healthy Alternatives:** Expects AI to recommend "swaps" that are healthier but remain within their price range.

Preferred Explanation Methods

1. **Visual Charts & Graphs:** Prefers color-coded visualizations that make data "visually pleasing" and easy to digest at a glance.
2. **Feature Importance Ranking:** A ranking of factors (e.g., "Why this product?") showing it was chosen because it's chemical-free and currently on sale.
3. **Confidence Scores:** Using clear thresholds to show how well a product matches their "No Artificial Sweeteners" filter.

How This Persona Guides Design Thinking

1. **UI Customization:** Directly influences the inclusion of **priority sliders** (Health vs. Eco) to ensure the interface is personalized to the user's specific values.
2. **Transparency Standards:** Supports the development of a **visual scoring system** that uses colors (e.g., Green to Red) to lower the cognitive load during daily shopping trips.
3. **Data Integration:** Highlights the need for a database that tracks **seasonal trends** and **ingredient-level data** (like artificial sweeteners) to provide the level of transparency the user demands.

Fig. 8. User Persona 5 – The Conscious Budgeter (Yun Mo).



Fig. 9. Empathy Map – The Eco-Beginner.



Fig. 10. Empathy Map – The Fitness Pragmatist.



Fig. 11. Empathy Map – The Value-Driven Traditionalist.



Jan Kuca Master's student in Software Engineering at the Faculty of Informatics, Masaryk University, Brno, Czech Republic. His interests include human-computer interaction, UI/UX design, and AI.



Stefan Sidlovsky Master's student in Computer Science at the Faculty of Informatics, Masaryk University, Brno, currently participating in an exchange program at the University of the Balearic Islands (UIB), Palma de Mallorca.



Anna Maria Chovancova Bachelor's student in Computer Science at the Faculty of Informatics, University of Zilina in Slovakia. Her interests include AI studies, human-computer interactions and coding.



Álvaro Velasco Sobrino He received the B.Sc. degree in Mathematics from the University of the Balearic Islands (UIB). He is currently pursuing the M.Sc. degree in Intelligent Systems at the same institution. His main research interests include data analysis and mathematical optimization.